2.2	**The sources, origins, physical and working properties of ferrous and non-ferrous metals and their social and ecological footprint**	To apply knowledge and understanding of the advantages, disadvantages and applications of the following materials, in order to be able to discriminate between them and select appropriately.
		2.2.1 Ferrous metals: a mild steel (in topic 1) b stainless steel (in topic 1) c cast iron (in topic 1) d high carbon steel e tungsten steel.
		2.2.2 Non-ferrous metals: a aluminium (in topic 1) b copper (in topic 1) c brass (in topic 1) d tin e 7000 series aluminium f titanium.
		2.2.3 Sources and origins – where ferrous and non-ferrous metals are resourced/manufactured and their geographical origin: a USA, Russia, Sweden – iron ore b China – steel c USA, France, Australia – bauxite (aluminium) d USA, Chile, Zambia, Russia – copper e Indonesia, China – tin.
		2.2.4 The physical characteristics of each ferrous and non-ferrous metal: a conductivity b magnetism c density.

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Key idea	What students need to learn:
	2.2.5 Working properties – the way in which each material behaves or responds to external sources: a ductility (in topic 1) b malleability (in topic 1) c hardness (in topic 1) d durability e toughness f elasticity g tensile strength h compressive strength.
	2.2.6 Social footprint: a trend forecasting b impact of extraction and material production on communities and wildlife c ease and difficulty of recycling and disposal.
	2.2.7 Ecological footprint: a sustainability b extraction and erosion of the landscape c processing d transportation e wastage f pollution.